

From Venus to Vividh Bharti: More on Damped, Driven Oscillations

1 An Experiment on Venus

We consider a mass m attached to a vertically suspended loose spring of spring constant k and equilibrium length l , placed in the dense atmosphere of Venus. Due to the atmosphere, the mass experiences a linear damping force

$$F_{\text{damp}} = -b\dot{x}.$$

Because the atmosphere of Venus is very dense so, the system is studied under the assumption of very strong damping.

(a) Equilibrium position

At equilibrium, the mass is at rest, so the damping force vanishes (as it depends upon velocity). The forces acting on the mass are gravity (mg , downward) and the spring restoring force (kx_0 , upward), where x_0 is the extension of the spring from its natural length.

Balancing the forces,

$$kx_0 = mg.$$

Hence, the new equilibrium position is

$$x_0 = \frac{mg}{k}.$$

(It is important to note that the gravitational acceleration g here refers to its value on Venus and not on Earth.)

(b) Equation of motion about equilibrium

Let $y(t)$ denote the displacement of the mass from the equilibrium position:

so, instantaneous position of mass;

$$x(t) = x_0 + y(t).$$

Applying Newton's second law,

$$m\ddot{x} = -kx - b\dot{x} + mg$$

$$m\ddot{x}_0 + m\ddot{y} = \cancel{kx_0} - ky - b\dot{x}_0 - b\dot{y} + \cancel{mg}$$

since $\dot{x}_0 = 0$ (and hence $\ddot{x} = 0$ also) we have ,

$$m\ddot{y} = -ky - b\dot{y}$$

which is equation of a damped harmonic oscillator, Thus, the equation of motion becomes

$$m\ddot{y} + b\dot{y} + ky = 0.$$

or,

$$\ddot{y} + \gamma\dot{y} + \omega_0^2 y = 0$$

The natural angular frequency of the system is

$$\omega_0 = \sqrt{\frac{k}{m}},$$

which is independent of the gravitational acceleration g . Gravity only determines the equilibrium position, not the oscillatory dynamics.

(c) Extremely overdamped limit

Defining

$$\gamma = \frac{b}{m}, \quad \omega_0^2 = \frac{k}{m},$$

as you may remember, the characteristic equation is

$$\alpha^2 + \gamma\alpha + \omega_0^2 = 0.$$

Its solutions are

$$\mu_1 = \frac{\gamma}{2} + \sqrt{\frac{\gamma^2}{4} - \omega_0^2},$$
$$\mu_2 = \frac{\gamma}{2} - \sqrt{\frac{\gamma^2}{4} - \omega_0^2}.$$

so the solution for overdamped oscillation looks like

$$x_{\text{overdamped}}(t) = C_1 e^{-\mu_1 t} + C_2 e^{-\mu_2 t}.$$

In the extremely overdamped limit ($\gamma \gg 2\omega_0$), we obtain the approximations

$$\mu_1 \approx \gamma,$$

and,

$$\mu_2 \approx \frac{\omega_0^2}{\gamma} = \frac{k}{b}.$$

(d) Dominant term at late times

The general solution for overdamped motion is

$$x(t) = C_1 e^{-\mu_1 t} + C_2 e^{-\mu_2 t}.$$

Since we have given a loose spring so k is "small" and we are in the dense atmosphere of Venus so b is "large" so from the above results of overdamping case $\mu_1 \gg \mu_2$, the term involving $e^{-\mu_1 t}$ decays rapidly. Therefore, μ_2 is important after few moments into the motion,

$$x(t) \approx C_2 e^{-\mu_2 t}.$$

(e) Qualitative description of the motion

As we have learnt in our class, in the case of overdamping there is not any oscillatory motion. Starting from rest at a finite displacement, the mass does not oscillate. Instead, it slowly and monotonically relaxes toward equilibrium. The motion is extremely sluggish, with no overshoot, (you can assume it like movement of mass in honey), which is characteristic of strong over damping (of course it is because of dense atmosphere of Venus)

(f) Force balance at late times

Using the late-time solution,

$$x(t) = C e^{-\mu_2 t},$$

the velocity is

$$\dot{x}(t) = -\mu_2 x(t).$$

The damping force is

$$F_{\text{damp}} = -b\dot{x} = b\mu_2 x,$$

while the spring force is

$$F_{\text{spring}} = -kx.$$

Substituting $\mu_2 = k/b$, we find

$$F_{\text{damp}} \approx -F_{\text{spring}}.$$

Thus, to leading order, the damping force balances the restoring force.

(g) Measuring the damping coefficient

From the late-time dynamics we have found that $\mu_2 \approx \frac{k}{b}$ so instantaneous velocity can be approximated as,

$$\dot{x} = -\frac{k}{b}x,$$

which can be rearranged to give

$$b = -\frac{kx}{\dot{x}}.$$

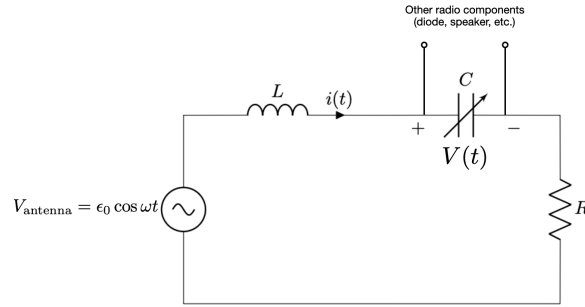
Therefore, by measuring the position and speed of the mass at a given time and using the known value of k , the damping coefficient(b) of Venus's atmosphere can be determined.

(h) Testing of calculations

After switching off the driving force and increasing the damping while keeping the tension small, we immediately notices a qualitative change in the system's behavior. The familiar oscillatory motion characterized by sine- or cosine-like variations completely disappears. Instead, the displacement decays smoothly and monotonically with time, following an exponential curve. This absence of oscillations is not surprising; it is precisely what we expects in the overdamped regime, where strong damping suppresses inertia and prevents the system from oscillating about its equilibrium position (Hope you have imagined similiar motion in section (e)).

2 Let's Tune In to the India-Pak Match

(a) Finding the constants in the differential equation



According to Kirchoff's voltage law, we can conclude that the sum of voltages across each component would be equal to $V(t)$. Now the voltage across each component is:

$$\frac{dV_C}{dt} = \frac{i}{C} \quad (1)$$

$$V_L = L \frac{di}{dt} = LC \ddot{V}_C \quad (2)$$

$$V_R = iR = RC \dot{V}_C \quad (3)$$

This gives,

$$V_L + V_R + V_C = \epsilon_0 \cos(\omega t) \quad (4)$$

$$LC \ddot{V} + RC \dot{V} + V = \epsilon_0 \cos(\omega t) \quad (5)$$

$$\ddot{V} + \frac{R}{L} \dot{V} + \frac{1}{LC} V = \frac{\epsilon_0}{LC} \cos(\omega t) \quad (6)$$

This thus gives,

$$\gamma = \frac{R}{L} \quad \omega_0^2 = \frac{1}{LC} \quad V_0 = \frac{\epsilon_0}{LC} \quad (7)$$

(b) Finding and analysing Voltage

Lets write the differential equation as the following for convenience:

$$\ddot{V} + \gamma\dot{V} + \omega_0^2 V = V_0 \cos \omega t. \quad (8)$$

We also know that the solution of such a differential equation has two parts, the homogeneous part and the particular part. Since γ is to be assumed to be small, we assume $\gamma/2 < \omega_0$. Thus the general solution for this equation would be:

$$V(t) = V_0 e^{-\frac{\gamma}{2}t} \cos(\beta t) + A \cos(\omega t - \phi) \quad (9)$$

Where the other parameters are:

$$\beta = \sqrt{\omega_0^2 - \left(\frac{\gamma}{2}\right)^2} \quad (10)$$

$$A = \frac{V_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2}} \quad (11)$$

$$\phi = \tan^{-1} \left(\frac{\gamma \omega}{\omega_0^2 - \omega^2} \right) \quad (12)$$

$$\omega = 2\pi f \quad (13)$$

At large t , the first term becomes negligible due to the exponential decay of the amplitude. Hence the amplitude is simply A (11).

(c) Finding the capacitance for tuning in

In this case, the natural frequency is changing, while the driving frequency remains constant. To find ω_0 such that the Amplitude of oscillation is maximum, we must maximize 11 with respect to ω_0 .

$$\frac{dA}{d\omega_0} = \frac{d}{d\omega_0} \left(\frac{\epsilon_0 \omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2}} \right) = 0 \quad (14)$$

$$\Rightarrow \frac{2\epsilon_0 \omega_0}{\sqrt{(\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2}} - \frac{2\omega_0^3 \epsilon_0 (\omega_0^2 - \omega^2)}{((\omega_0^2 - \omega^2)^2 + \omega^2 \gamma^2)^{\frac{3}{2}}} = 0 \quad (15)$$

$$\Rightarrow \omega_0^2 = \omega^2 + \gamma^2 \quad (16)$$

Here γ is really small compared to ω_0 , and thus one can approximate $\omega_0 = \omega$. The capacitance to tune into the radio station can be found by using this condition. We know that $\omega_0 = \frac{1}{\sqrt{LC}}$, and $\omega = 2\pi f$. Equating these two gives:

$$C = \frac{1}{4\pi^2 f^2 L} \quad (17)$$

3 Interference Tolerance

(a) Finding the width of the response peak

Here we have the freedom of adjusting the natural frequency unlike the regular case of forced damped oscillations, where we have the freedom to choose the frequency of the driving force. Thus the amplitude is maximum at $\omega_0 = \omega$. The maximum amplitude is thus:

$$A_{max} = \frac{V_0}{\omega \gamma} \quad (18)$$

Now to find the maximum range of ω_0 such that the maximum amplitude of the voltage is no less than αA_{max} , we equate the following quantities:

$$(\alpha A_{max})^2 = \frac{V_0^2}{(\omega_0'^2 - \omega^2)^2 + \omega^2 \gamma^2} \quad (19)$$

$$(20)$$

Near resonance, $\omega = \omega_0 + \Delta$. So,

$$\omega_0^2 - \omega^2 = -2\Delta\omega_0 \quad (21)$$

Thus,

$$\left(\frac{V_0}{\omega\gamma}\alpha\right)^2 = \frac{V_0^2}{\omega^2(4\Delta^2 + \gamma^2)} \quad (22)$$

$$\Rightarrow \Delta = \pm \frac{\gamma}{2} \sqrt{\frac{1}{\alpha^2} - 1} \quad (23)$$

This is the range of the natural frequency.

(b) Voltage across capacitor including interference

Due to the interference due to the nearby radio station, the differential equation becomes:

$$\ddot{V} + \gamma\dot{V} + \omega_0^2 V = V_0 \cos \omega t + V_0 \cos \omega' t \quad (24)$$

where $\omega' = 2\pi f$. This differential equation would have two particular solutions as opposed to one in the previous case. This is due to two driving forces. Summing up these solutions linearly would give us a superposition of the effects of the two driving forces. Thus the general solution for the voltage across the capacitance would now be:

$$V(t) = V_0 e^{-\frac{\gamma}{2}t} \cos(\beta t) + A \cos(\omega t - \phi) + A' \cos(\omega' t - \phi') \quad (25)$$

A' and ϕ' have their usual meaning.

(c) Finding the quality factor

To ensure that the interference remains minimum, we must first find an upper limit for γ . We do this by utilizing the condition $A'/A \leq 0.01$. Assume $k = 1.002$ and $c = 0.01$. Moreover, the capacitance must be such that $\omega_0 = \omega = 2\pi f$. We know that:

$$A = \frac{V_0}{\omega_0 \gamma} \quad (26)$$

Taking the ratio of the amplitudes of voltage from both the stations:

$$\frac{A'}{A} = \frac{1}{\sqrt{\frac{\omega_0^2(k^2-1)^2}{\gamma^2} + k^2}} \leq c \quad (27)$$

$$\Rightarrow \frac{\omega_0^2(k^2-1)^2}{\gamma^2} + k^2 \leq \frac{1}{c^2} \quad (28)$$

$$\Rightarrow \frac{1}{\gamma^2} \leq \left(\frac{1}{c^2} - k^2\right) \frac{1}{\omega_0^2(k^2-1)^2} \quad (29)$$

$$\Rightarrow \gamma \leq \frac{\omega_0(k^2-1)}{\sqrt{\frac{1}{c^2} - k^2}} \quad (30)$$

At resonance $\omega_0 = 2\pi f$. So:

$$Q = \frac{\omega_0}{\gamma} = \frac{\sqrt{\frac{1}{c^2} - k^2}}{(k^2-1)} = 25000 \quad (31)$$

Now to find the maximum resistance, we use the following relations:

$$\omega_0^2 = \frac{1}{LC} \quad \Rightarrow \quad L = \frac{1}{\omega_0^2 C} \quad (32)$$

$$\gamma = \frac{R}{L} = RC\omega_0^2 \quad (33)$$

$$(34)$$

Now using 30, we find:

$$R \leq \frac{1}{C\omega_0} \frac{(k^2-1)}{\sqrt{\frac{1}{c^2} - k^2}} \quad (35)$$